

Problem 1

1. On the Gisette dataset, train decision trees of maximum depth $1, 2, \dots$ up to 12, for a total of 12 decision trees. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test misclassification errors vs tree depth as two separate curves. Report in a table the minimum test error and the tree depth for which the minimum was attained.

We do this in Python, and obtain the following plot of the training and test misclassification errors vs tree depth:

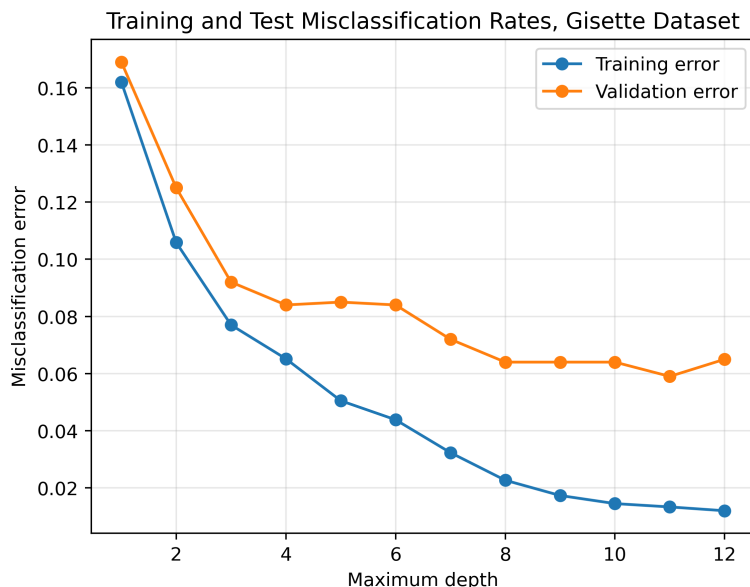


Figure 1: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.
We can do this also, and obtain the four images shown in figure 2.
3. On the Gisette dataset, for each off $k \in \{3, 10, 30, 100, 300\}$ train a random forest with k trees where the split attribute at each node is chosen from a random subset of $71 \approx \sqrt{5000}$ features. Use the trained trrees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test errors vs. number of trees k as two separate curves. Report the training and test misclassification errors in a table.
4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.
5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of $12 \approx \log_2(5000)$ features, and where the split attribute at each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.

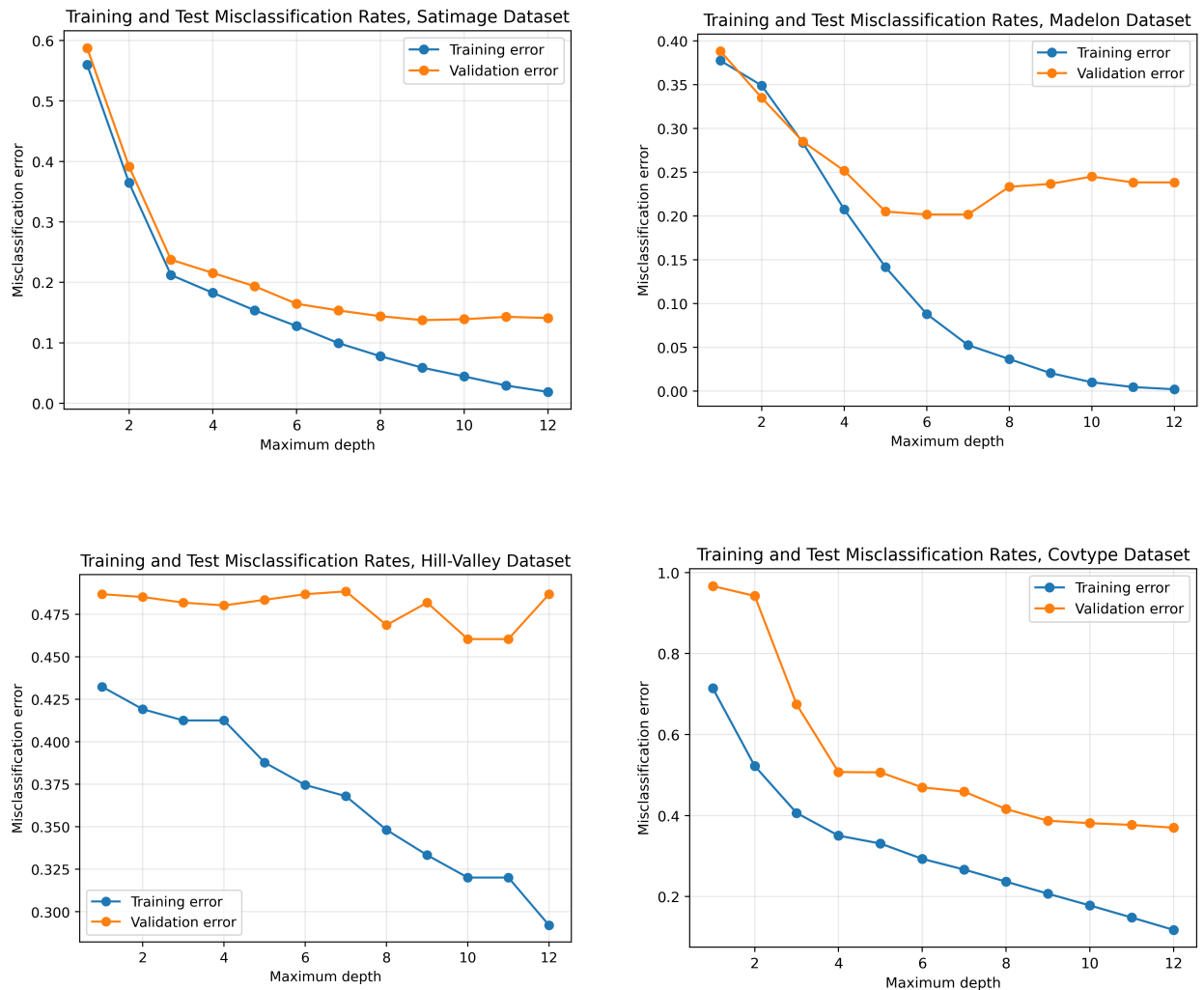


Figure 2: Training and Validation datasets misclassification error rates for four datasets